

Ebola Surveillance Dashboard — Client-Side Web App

Product Requirements Document • v1.0

Status: Approved

Owner: Product / DX Team

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Stack decision: Vanilla JS + Chart.js + HTML/CSS — zero build step, zero backend

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1. Executive Summary

Deliver a fully **client-side, single-file** epidemiological surveillance dashboard covering the 2014–2016 West Africa Ebola outbreak. The dashboard must open directly in any modern browser with no installation, no server, no API keys, and no build step — yet present data at the standard of a professional WHO/CDC situation room.

One deliverable: `index.html` — self-contained, embeds all data, ready to open.

2. Background & Context

A React + Recharts dashboard was previously built and works well inside Claude's artifact viewer. It cannot, however, be opened as a standalone file, distributed via email, or hosted on a static file server without a Node.js build pipeline.

This PRD specifies a ground-up rewrite using the browser's native capabilities and CDN-loaded Chart.js — producing the same analytical value with zero infrastructure requirements.

Previous Implementation Gaps

Gap	Impact
Requires React build toolchain (npm, bundler)	Cannot share as a file attachment
JSX syntax requires Babel compilation	Fails if opened directly as HTML
No offline capability	Unusable without internet after first load
Multiple component files	Requires a dev server to avoid CORS on <code>import</code>

3. Problem Statement

Public health data from the largest Ebola outbreak in history is scattered across PDF situation reports. Analysts, educators, journalists, and students need an interactive, shareable, always-available dashboard that:

- Requires **no technical setup** to run
 - Can be **emailed as a single file** and opened immediately
 - Presents **professional-grade visualisations** aligned to WHO reporting standards
 - Works **offline** once the CDN fonts/libraries are cached
-

4. Goals

#	Goal	Measure
G1	Zero-dependency runtime	No npm, no server, no API call to open the app
G2	Single-file deliverable	Entire app in one <code>index.html</code> ≤ 120 KB
G3	Full data coverage	All 6 dashboard sections from the original spec
G4	Animated, interactive	Counters animate on load; charts are interactive
G5	Consistent design language	Matches the dark-ops design system used across all tools
G6	Responsive	Readable at 375 px (mobile) through 1920 px (desktop)
G7	Accessible	WCAG 2.1 AA contrast, keyboard-navigable, screen-reader labels

5. Non-Goals

- **Real-time data** — all data is static, embedded at build time
- **User authentication** — no login, no accounts
- **Data entry or editing** — read-only; no forms that write to storage
- **PDF export** — out of scope for v1.0
- **Multilingual support** — English only for v1.0
- **Internet Explorer / legacy browser support** — modern browsers only (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)

6. User Personas

P1 — Public Health Educator

"I teach epidemiology at university. I want something I can share with students as a single file and have them explore independently."

- Needs: working app with no setup, clear labelling, educational context
- Device: laptop, projector during class
- Technical level: intermediate

P2 — Journalist / Science Communicator

- "I'm writing a retrospective on the outbreak. I need accurate figures quickly."
- Needs: instant access to verified numbers, fast load, readable on mobile
 - Device: phone, tablet, laptop
 - Technical level: low

P3 — Data / BI Analyst

- "I want to explore the district-level data and understand geographic hotspots."
- Needs: sortable table, CFR comparisons, tooltips with precise values
 - Device: desktop with large monitor
 - Technical level: high

P4 — Policy Researcher

- "I'm comparing intervention timing across outbreaks. I need the key event dates and case trajectory in one place."
- Needs: timeline, epidemic curve, country breakdown
 - Device: laptop
 - Technical level: medium

7. User Stories

ID	As a...	I want to...	So that...
US-01	Educator	Open the file by double-clicking it	Students can start immediately
US-02	Educator	See animated counters on load	The scale of the outbreak is immediately felt
US-03	Analyst	Toggle the epidemic curve between monthly new and cumulative	I can analyse both trend and total
US-04	Analyst	Hover over chart data points	I can read precise values for any month

ID	As a...	I want to...	So that...
US-05	Analyst	Sort the district table by cases, deaths, or CFR	I can rank districts by any metric
US-06	Journalist	See the overall CFR at a glance	I can cite the figure accurately
US-07	Journalist	See each country's CFR separately	I can compare Guinea vs Liberia vs Sierra Leone
US-08	Researcher	Read the key events timeline	I understand the sequence of the outbreak
US-09	Any user	Use the dashboard on a phone	I can access it from any device
US-10	Any user	Know where the data came from	I can trust and cite the figures

8. Functional Requirements

FR-01 • Header

ID	Requirement
FR-01.1	Display project title: EBOLA SURVEILLANCE DASHBOARD
FR-01.2	Display sub-title: West Africa · 2014-2016 · WHO / CDC Data
FR-01.3	Show a blinking green status indicator labelled OUTBREAK ENDED
FR-01.4	Display epidemic period: DEC 2013 — JUN 2016
FR-01.5	Header is sticky — remains visible on scroll

FR-02 • KPI Cards

ID	Requirement
FR-02.1	Display four KPI cards: Total Cases, Total Deaths, HCW Infections, Countries Affected
FR-02.2	Each card value animates from 0 to its target on first load using eased counter animation
FR-02.3	Sub-label below each value provides context (e.g. 39.5% CFR , 3 epicenters)
FR-02.4	Values: 28,616 cases · 11,310 deaths · 886 HCW infections · 10 countries
FR-02.5	Each card has a distinct accent colour matching the design system

FR-03 • Epidemic Curve

ID	Requirement
FR-03.1	Display an area chart of Ebola cases and deaths from Mar 2014 to Mar 2016 (25 months)
FR-03.2	Toggle between two views: Monthly New (default) and Cumulative
FR-03.3	Cases plotted in orange; deaths plotted in red; both as filled areas with transparency
FR-03.4	A dashed vertical reference line marks the WHO PHEIC declaration (Aug 2014)
FR-03.5	A dashed vertical reference line marks the epidemic peak (Oct 2014)
FR-03.6	Hovering any data point shows a dark-themed tooltip with month, cases, and deaths
FR-03.7	Y-axis labels abbreviate thousands: 1k , 5k , 8k
FR-03.8	Chart is fully responsive — redraws correctly on window resize

FR-04 • Country Distribution

ID	Requirement
FR-04.1	Display a doughnut chart showing case share by country
FR-04.2	Show Sierra Leone (49.4%), Liberia (37.3%), Guinea (13.3%), Others (<1%)

ID	Requirement
FR-04.3	Custom legend below the chart with country name, colour, percentage
FR-04.4	Hovering a segment shows country name and case count in a tooltip

FR-05 • Country Comparison Bar Chart

ID	Requirement
FR-05.1	Display a horizontal grouped bar chart with cases and deaths for each country
FR-05.2	Include the five affected countries: Sierra Leone, Liberia, Guinea, Nigeria, Mali
FR-05.3	Cases bar is orange; deaths bar is red
FR-05.4	Y-axis shows country names; X-axis shows count (abbreviated)

FR-06 • CFR Analysis

ID	Requirement
FR-06.1	Display a CSS progress-bar section showing CFR for each of the five main countries
FR-06.2	Bar colour encodes severity: CFR > 60% = red, > 40% = orange, else = yellow
FR-06.3	CFR percentage is shown as a numeric label to the right of each bar
FR-06.4	A summary line at the bottom shows the overall average CFR: 39.5%

FR-07 • District Hotspot Table

ID	Requirement
FR-07.1	Display the top 15 most-affected districts in a sortable table

ID	Requirement
FR-07.2	Columns: Rank · District · Country · Cases · Deaths · CFR · Severity
FR-07.3	Clicking a column header sorts the table by that column (toggle asc/desc)
FR-07.4	Severity badge has four levels: EXTREME (CFR > 60%), HIGH (> 40%), MODERATE (> 25%), ELEVATED
FR-07.5	Alternating row background for readability
FR-07.6	Country column uses a colour-coded badge matching the design system

FR-08 · Key Events Timeline

ID	Requirement
FR-08.1	Display a vertical timeline of 11 key outbreak milestones
FR-08.2	Each entry shows: date, event description, dot coloured by type
FR-08.3	Dot colours: red = critical, green = positive milestone, yellow = alert, cyan = end
FR-08.4	A vertical spine connects all dots
FR-08.5	The HCW callout card appears below the timeline entries

FR-09 · Footer

ID	Requirement
FR-09.1	Display data sources: WHO · CDC · HDX · NEJM
FR-09.2	Display the data accuracy disclaimer
FR-09.3	Display generation date and license

9. Non-Functional Requirements

Performance

ID	Requirement
NFR-01	Time to interactive: < 2 seconds on a 10 Mbps connection (first load)
NFR-02	Time to interactive: < 200 ms on subsequent loads (cached CDN assets)
NFR-03	Total HTML file size: ≤ 120 KB uncompressed
NFR-04	No layout shift or flash of unstyled content on load

Compatibility

ID	Requirement
NFR-05	Works in Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
NFR-06	Works at viewport widths from 375 px to 1920 px
NFR-07	Opens directly from the filesystem (file:// protocol) without errors
NFR-08	No service worker, no localStorage, no cookies

Reliability

ID	Requirement
NFR-09	If Google Fonts CDN fails, app falls back to system fonts
NFR-10	If Chart.js CDN fails, a visible error message replaces each chart
NFR-11	All data is embedded; charts render even with no internet after CDN loads

Accessibility

ID	Requirement
NFR-12	All colour combinations pass WCAG 2.1 AA contrast (4.5:1 normal, 3:1 large text)
NFR-13	All interactive elements (toggles, sortable headers) are keyboard-operable
NFR-14	Chart canvases have <code>aria-label</code> describing the chart content
NFR-15	The district table uses proper <code><thead></code> , <code><th scope></code> , <code><tbody></code> markup

10. Data Model

All data is embedded as JavaScript constants in the `<script>` block. No external data fetches.

```
MONTHS[25]           - Month labels, Mar '14 through Mar '16
MCASES[25]           - Monthly new cases (aggregate, all countries)
MDEATHS[25]          - Monthly new deaths
MCUM_CASES[25]       - Cumulative cases (derived from MCASES)
MCUM_DEATHS[25]      - Cumulative deaths (derived from MDEATHS)
COUNTRIES[5]         - {name, flag, cases, deaths, cfr, color}
DISTRICTS[15]        - {name, country, iso, cases, deaths, cfr}
KEY_EVENTS[11]       - {date, text, type}
PIE_DATA[4]          - {name, value, color} - country case share
```

Data Sources & Accuracy

Dataset	Primary Source	Notes
Monthly aggregates	WHO Ebola Situation Reports	Approximate — reconstructed from published reports
Country totals	WHO Final Report (Jun 2016)	Exact figures
District totals	WHO AFRO + HDX	Approximate
HCW data	WHO HCW Situation Reports	Approximate
Key event dates	CDC MMWR, WHO DoN	Exact

11. UI/UX Specifications

Layout Grid

HEADER (sticky, 58px)				
KPI CARD	KPI CARD	KPI CARD	KPI CARD	← 4 × 25% col
EPIDEMIC CURVE (area chart)		COUNTRY PIE (donut + legend)		← 65% / 35%
COUNTRY BAR CHART (grouped)		CFR ANALYSIS (CSS progress)		← 65% / 35%
DISTRICT TABLE (sortable)		KEY EVENTS (timeline)		← 65% / 35%
FOOTER				

Responsive Breakpoints

Breakpoint	Layout
> 1024px	Full grid as above
768–1024px	KPI 2×2 grid; chart pairs stack vertically
< 768px	Single column; all cards full width

Interaction States

Element	Default	Hover	Active	Focus
Toggle button	muted border	highlight border	accent bg	visible ring
Table header	arrow icon (↑)	colour change	sort applied	visible ring

Element	Default	Hover	Active	Focus
Chart tooltip	hidden	visible	—	—

12. Component Inventory

Component	Type	Chart.js Type	Notes
Header	HTML/CSS	—	Sticky, contains status pill
KPI Card × 4	HTML/CSS/JS	—	Animated counter on load
Epidemic Curve	Chart.js	line + fill	Toggle: new / cumulative
Country Donut	Chart.js	doughnut	Custom legend
Country Bars	Chart.js	bar (horizontal)	Grouped: cases + deaths
CFR Bars	HTML/CSS	—	CSS width = CFR%
District Table	HTML/CSS/JS	—	Client-side sort
Key Events	HTML/CSS	—	Vertical timeline
Footer	HTML/CSS	—	Sources + disclaimer

13. Design System Tokens

Colours

Token	Hex	Usage
--bg	#04070f	Page background
--surface	#080d1e	Header, section backgrounds
--card	#0b1229	Card backgrounds
--border	#152040	Card borders, dividers
--orange	#f97316	Cases, primary accent

Token	Hex	Usage
--red	 #ef4444	Deaths, critical severity
--green	 #22c55e	Positive milestones, resolved
--cyan	 #22d3ee	Informational, end-of-outbreak
--yellow	 #fbbf24	Alerts, HCW data
--purple	 #c084fc	CFR, Liberia
--text	 #e2e8f0	Primary text
--mid	 #94a3b8	Secondary text
--muted	 #475569	Labels, captions

Typography

Token	Value	Usage
--display	Barlow Condensed	Headings, KPI values, section titles
--mono	IBM Plex Mono	Data labels, captions, code, badges
Body	system-ui	Descriptive text

Spacing Scale

4 · 8 · 12 · 16 · 20 · 24 · 32 · 40 · 48 px

Border Radius

6px (small elements), 10px (cards), 12px (large cards), 20px (pills)

14. Technical Architecture

```
ebola-dashboard-v2/
├─ index.html           ← single deliverable; everything inside
  │ └─ <head>
  │   │ └─ meta, viewport, title
  │   │ └─ Google Fonts (Barlow Condensed + IBM Plex Mono)
```

```

|   └─ Chart.js 4.x via jsDelivr CDN
|
| └─ <style>
|   └─ CSS custom properties (design tokens)
|   └─ Reset + base styles
|   └─ Layout (CSS grid, flexbox)
|   └─ Component styles (header, cards, table, timeline)
|   └─ Animations (blink, fade-up, counter)
|   └─ Responsive media queries
|
| └─ <body>
|   └─ #header
|   └─ #kpi-row (4 × .kpi-card)
|   └─ #row-1 (.chart-card × 2)
|   └─ #row-2 (.chart-card × 2)
|   └─ #row-3 (.data-card × 2)
|   └─ <footer>
|
| └─ <script>
|   └─ 1. DATA CONSTANTS
|   └─ 2. CHART.JS GLOBAL DEFAULTS
|   └─ 3. REFERENCE LINE PLUGIN (inline, no extra CDN)
|   └─ 4. KPI COUNTER ANIMATION
|   └─ 5. EPIDEMIC CURVE – init + toggle handler
|   └─ 6. COUNTRY DONUT – init
|   └─ 7. COUNTRY BAR CHART – init
|   └─ 8. CFR BARS – DOM build
|   └─ 9. DISTRICT TABLE – DOM build + sort
|   └─ 10. KEY EVENTS TIMELINE – DOM build
|   └─ 11. DOMContentLoaded → run all inits

```

CDN Dependencies

Library	Version	URL	Purpose
Chart.js	4.4.0	jsDelivr	All chart rendering
Google Fonts	—	fonts.googleapis.com	Barlow Condensed, IBM Plex Mono

Fallback policy: If Chart.js fails to load, each canvas is replaced with an error message. If Google Fonts fails, the app falls back to `system-ui` and `'Courier New'` respectively.

15. File Structure

```
ebola-dashboard-v2/  
└─ index.html      ← The entire application. Nothing else required.
```

This is intentional. The goal is a single file that can be:

- Emailed as an attachment
 - Opened from a USB stick
 - Hosted on any static file server (GitHub Pages, S3, Netlify)
 - Shared via Slack / Teams without any setup instructions
-

16. Acceptance Criteria

Must Pass Before Ship

- ☐ Opens directly from filesystem (`file://`) in Chrome, Firefox, Safari — no server required
- ☐ All four KPI counters animate from 0 to target on load
- ☐ Epidemic curve renders with both Monthly New and Cumulative toggle states
- ☐ Both reference lines (PHEIC Aug 2014, Peak Oct 2014) are visible on the curve
- ☐ Country donut renders with correct percentages (SLE 49.4%, LBR 37.3%, GIN 13.3%)
- ☐ Country bar chart renders with grouped cases + deaths for all five countries
- ☐ CFR bars display correct values and correct severity colours
- ☐ District table renders all 15 rows and sorts correctly on all three columns
- ☐ Key events timeline renders all 11 milestones with correct dot colours
- ☐ Tooltip appears on chart hover showing precise values
- ☐ No console errors on load
- ☐ Layout is readable at 375 px viewport width
- ☐ File size is $\leq$ 120 KB

Should Pass

- ☐ `aria-label` on all chart `<canvas>` elements
 - ☐ Table uses correct `<th scope="col">` and `<th scope="row">` attributes
 - ☐ Keyboard navigation reaches all interactive controls
 - ☐ All text passes WCAG AA contrast ratio
-

17. Success Metrics

Metric	Target
File opens without a server	✔ Pass
Load time (cached CDN)	< 200 ms
File size	≤ 120 KB
Chart render errors	0
Console errors on load	0
Responsive at 375 px	✔ Pass

18. Open Questions

#	Question	Owner	Status
OQ-1	Should sorting persist between page refreshes (localStorage)?	Product	Deferred to v1.1
OQ-2	Should we add a print stylesheet so the dashboard prints cleanly?	Design	Deferred to v1.1
OQ-3	Is there a requirement to support right-to-left languages?	Product	No — English only in v1.0
OQ-4	Should the toggle state (new vs cumulative) persist on refresh?	Engineering	No — always reset to Monthly New

*This PRD is the canonical specification for `ebola-dashboard-v2/index.html`.
All implementation decisions should trace back to a requirement listed here.
Changes require a version bump and changelog entry.*